

## **Clinic Management System – Digital Healthcare & Clinical Workflow PWA**

### **Project Overview**

The Clinic Management System is an end-to-end Progressive Web Application (PWA) designed to streamline clinical workflows, patient registrations, appointment scheduling, doctor consultations, electronic health records (EHR), e-prescriptions, lab tests, and billing. The platform connects receptionists, doctors, lab technicians, cashiers, and patients into a unified digital ecosystem that eliminates paper medical files and reduces patient wait times.

Built with an offline-first architecture, the application allows doctors and clinical staff to access patient medical histories and log consultation notes seamlessly, automatically synchronizing data once an internet connection is available.

### **Problem Statement**

Many private clinics and diagnostic centers in Ethiopia rely on physical paper charts, hand-written prescription pads, and manual queue systems. This approach causes major operational challenges:

**Lost or Disorganized Medical Files:** Physical paper folders are easily misplaced, taking significant time to retrieve and preventing doctors from viewing complete patient histories.

**Long Waiting Times & Chaotic Queues:** Without digital scheduling, patients experience long, unpredictable wait times in reception areas, leading to frustration and crowded waiting rooms.

**Unreadable Hand-written Prescriptions:** Illegible prescription handwriting increases the risk of dispensing errors at pharmacies and delays patient treatment.

**Fragmented Billing & Revenue Leakage:** Manual cross-checking between doctor consultations, lab test orders, and cashier desks frequently results in unbilled services and unaccounted clinic revenue.

**Disconnected Diagnostic Results:** Lab technicians produce physical test reports that must be manually carried back to doctors, slowing down critical medical decisions.

This project solves these real-world healthcare bottlenecks by providing a fast, secure, and coordinated digital workflow.

### **Proposed Solution**

Develop a high-performance Progressive Web Application (PWA) that enables clinics to:

Register patients with unique Medical Record Numbers (MRN) and digital health profiles.

Manage doctor shift schedules, time-slot bookings, and real-time patient queue order.

Provide doctors with a structured Electronic Health Record (EHR) interface for logging SOAP notes (Subjective, Objective, Assessment, Plan) and diagnostic codes.

Generate digital e-prescriptions with automatic dosage calculations and printable PDF downloads.

Digitally dispatch laboratory test orders to the clinic's diagnostic department and receive results in real-time.

Process itemized billing invoices for consultations, procedures, and lab services with thermal receipt printing.

### **Target Users**

The platform supports multi-role interactions within outpatient clinics:

Clinic Administrators & Directors

Receptionists & Triage Nurses

Doctors & General Practitioners

Laboratory Technicians

Cashiers & Finance Officers

Patients

## **Objectives**

The project aims to:

Transition clinics to 100% paperless electronic health record keeping.

Minimize patient wait times through structured time-slot booking and digital queue management.

Prevent medication errors by generating clear digital prescriptions.

Unify billing for consultations, lab tests, and treatments to ensure total revenue accountability.

Accelerate laboratory report turnaround times through instant digital notification workflows.

## **Functional Requirements**

### **Patient Registration & Triage**

Search, onboard, and manage patient profiles with unique Medical Record Numbers (MRN).

Record emergency contacts, allergies, blood group, and chronic conditions.

Nurse triage module: Record initial vital signs (blood pressure, temperature, pulse, weight, BMI).

### **Doctor Schedules & Appointment Booking**

Define weekly doctor availability, shift slots, and consultation durations.

Book walk-in and advance appointments with live queue numbers.

Real-time queue status tracker for receptionists and waiting area monitors.

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### **Desk**

Doctor clinical desk with complete access to past consultation notes, vitals history, and lab reports.

Structured SOAP notes entry form (Chief Complaints, Physical Examination, Diagnosis, Treatment Plan).

Diagnostic lookup (ICD-10 standard codes integration).

### **Digital E-Prescriptions**

E-prescription generator with dosage, frequency, route, and duration fields.

Auto-generate official printable prescription PDFs with doctor digital signatures and clinic letterheads.

Direct handoff interface for internal or partner pharmacy fulfillment.

### **Laboratory & Diagnostics Management**

Doctor lab order creation during consultation.

Lab technician portal showing pending test queues, sample collection status, and result entry fields.

Attach diagnostic file uploads or auto-generate official lab test report PDFs.

### **Cashier Desk & Itemized Billing**

Combined invoice generation for consultation fees, lab procedures, and medical supplies.

Payment status tracking (Pending, Paid, Partial) across Cash, Mobile Money (Telebirr, CBE Birr), and Insurance.

Generate thermal payment receipts and formal medical claim invoices.

## **Analytics & Operational Reports**

Daily financial revenue summary split by service type (Consultation, Lab, Procedures).

Patient flow analytics: Average wait times, daily consultation volume, peak clinic hours.

Disease incidence and diagnostic distribution reports.

## **Non-Functional Requirements**

**Progressive Web Application (PWA):** Installable across desktop PCs, tablets, and mobile devices.

**Offline-First Capabilities:** Local browser storage for recording vital signs and consultation notes during internet disruptions.

**Data Security & Privacy:** HIPAA-compliant standards for Patient Health Information (PHI) encryption at rest and in transit.

**Role-Based Security:** Strict access separation ensuring lab technicians and cashiers only see data relevant to their specific tasks.

**Fast Response Time:** High-speed dashboard loading (<200ms) for high-volume clinic reception desks.

**Localization Support:** Multi-language interface support (English, Amharic, Afaan Oromo).

## **Technical Architecture & Preferred Tech Stack**

### **Frontend Options**

**TanStack Start (Preferred)** + React + Tailwind CSS + Shadcn UI

**React.js (SPA)** with Vite + Tailwind CSS

### **Backend Options**

**NestJS (Preferred for enterprise scale)** (Modular REST API with WebSocket queue updates)

**Node.js + Express.js** (REST API)

**Golang** (High-concurrency API)

### **Database & ORM Options**

**PostgreSQL** (Primary Relational Database)

**Drizzle ORM (Preferred) / Prisma ORM**

### **Authentication & Authorization**

**JWT (JSON Web Tokens)** with Refresh Token rotation.

**RBAC Engine** enforcing Admin, Doctor, Nurse, Receptionist, Lab Tech, and Cashier access boundaries.

### **Future Enhancements**

**Telehealth / Video Consultations:** Virtual doctor appointments integrated with video calling.

**Patient Self-Service Portal & Mobile App:** Patient portal for downloading lab results, viewing e-prescriptions, and booking appointments remotely.

**Insurance Claim Integration:** Direct integration with third-party medical insurance approval portals.

**Automated SMS Appointment Reminders:** Automated SMS alerts sent 24 hours prior to scheduled consultation times.

### **Business Value**

The Clinic Management System PWA replaces paper files and scattered clinic desks with an integrated, high-speed digital healthcare platform. By eliminating file retrieval delays, streamlining lab workflows, preventing billing oversights, and speeding up doctor

consultations, the platform improves patient care quality while driving higher daily patient throughput and financial performance for clinic owners.